

Countering A2/AD Through EABO:
The Role and Feasibility of Distributed ISR in the Indo-Pacific



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ABSTRACT

The potential for conflict in the Indo-Pacific is rapidly growing as the People's Republic of China (PRC) consistently increases its military exercises in the vicinity of Taiwan and its harassment of Philippine vessels in the South China Sea. The United States and its allies are preparing how they will respond in the event one of these flashpoints escalates.

The U.S. Navy has operated consistently around the globe for decades for the purpose of maintaining presence and supporting its power projection. In recent years, as the PRC has grown more aggressive in its territorial claims, the U.S. Navy has proudly conveyed that its presence in China's backyard is to support a "Free and Open Indo-Pacific."

The PRC has a wide array of systems in its anti-access/area denial (A2/AD) arsenal specifically designed to counter that idea. A review of this potential battlespace shows a PRC military with an extensive A2/AD complex that aims to keep the U.S. and its allies from interfering with its interests. The U.S. military is adjusting and developing its procedures, tactics, and operations to respond to this threat. In order to achieve its objectives, the U.S. will need to break through the A2/AD wall that the PRC has built and operate within this contested area.

An evolving military strategy to support this objective is the concept of using small, low-footprint, low-signature units dispersed through the critical archipelagic islands of the First Island Chain to get U.S. and allied forces closer to critical maritime terrain without exposing their more valuable assets to the threat of the PRC's ballistic missile arsenal. A core tenet of this concept of dispersed operations is called Expeditionary Advanced Base Operations (EABO).

EABO has potential to increase the situational awareness of U.S. forces and provide critical information to fleet and combatant commanders using dispersed intelligence, surveillance, and reconnaissance (ISR) capabilities. This concept, however, faces serious challenges. Placing U.S. and allied forces in these locations, within the weapons engagement zone (WEZ) of PRC systems, could certainly expose them to detection and engagement by adversary forces. In addition, they will require robust and resilient communications to ensure the information they gather can be properly disseminated in this contested environment.

In addition to the operational and tactical challenges posed by this strategy, there are potential geopolitical barriers that could prevent the employment of EABO. The critical littoral region of the First Island Chain, where EABO units would operate, is primarily territory of the Philippines and Japan. These host nations may have conflicting goals regarding the escalation of any conflict in the region. Any authority granted by host nations for EABO units to operate on their terrain could be met with retaliation from the PRC. For the purpose of this study, access to this terrain is assumed to be granted. Although access cannot be guaranteed and host nation political will can be variable, this assumption allows for a thorough evaluation of the tactical and operational feasibility of the concept, independent of diplomatic fluctuations.

This research assesses the feasibility of EABO as a tactical concept and determines if the capabilities it provides are practical. In order to evaluate the feasibility and practicality of this concept, a thorough review of the potential systems it would employ is included. In addition, the specific tactics that would be employed to ensure survivability are discussed. It explores the necessity of resilient communications for data sharing and the critical need for signal management to avoid detection. In conclusion, the analysis discusses the potential strengths of the concept as well as the critical vulnerabilities that could prevent it from achieving strategic success.

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LIST OF ABBREVIATIONS

A2/AD	Anti-Access/Area Denial
ACE	Agile Combat Employment
AI	Artificial Intelligence
ASBM	Anti-Ship Ballistic Missile
ASCM	Anti-Ship Cruise Missile
BDA	Battle Damage Assessment
BLoS	Beyond-Line-of-Sight
C2	Command and Control
C5ISR-T	Command, Control, Communications, Computers, Cyber, Intelligence, Surveillance, Reconnaissance, Targeting
CAC2S	Common Aviation Command and Control System
CEC	Cooperative Engagement Capability
CJADC2	Combined Joint All-Domain Command and Control
COP	Common Operational Picture
CTP	Common Tactical Picture
DDG	Guided Missile Destroyer
DMO	Distributed Maritime Operations
DOTMLPF-P	Doctrine, Organization, Training, Materiel, Leadership, Personnel, Facilities, Policy
DSSS	Direct Sequence Spread Spectrum
EAB	Expeditionary Advanced Base
EABO	Expeditionary Advanced Base Operations
ELINT	Electronic Intelligence
EM	Electromagnetic
EMCON	Emissions Control
EO	Electro-Optical
EO/IR	Electro-Optical/Infrared
ESM	Electronic Support Measures
EW	Electronic Warfare
F2T2EA	Find, Fix, Track, Target, Engage, and Assess
FD2030	Force Design 2030
FHSS	Frequency Hopping Spread Spectrum
G/ATOR	Ground/Air Task Oriented Radar
GARC	Global Autonomous Reconnaissance Craft
GPS	Global Positioning System
HF	High-Frequency
HGV	Hypersonic Glide Vehicle
HIMARS	High Mobility Artillery Rocket System
IADS	Integrated Air Defense System
IAMD	Integrated Air and Missile Defense
IIR	Imaging Infrared
IR	Infrared
IRBM	Intermediate-Range Ballistic Missile
ISR	Intelligence, Surveillance, and Reconnaissance
JADC2	Joint All-Domain Command and Control
JLTV	Joint Light Tactical Vehicle
LAAB	Littoral Anti-Air Battalion
LCT	Littoral Combat Team

LEO	Low Earth Orbit
LLB	Littoral Logistics Battalion
LOCE	Littoral Operations in a Contested Environment
LOS	Line-of-Sight
LPD	Low Probability of Detection
LPI	Low Probability of Intercept
LRASM	Long-Range Anti-Ship Missile
LRFL	Long Range Fires Launch
LSM	Landing Ship Medium
MAGTF	Marine Air Ground Task Force
MANGL	MAGTF Agile Network Gateway Link
ML	Machine Learning
MLR	Marine Littoral Regiment
MRBM	Medium-Range Ballistic Missile
MST	Maritime Strike Tomahawk
NIFC-CA	Naval Integrated Fire Control – Counter Air
NMESIS	Naval/Marine Expeditionary Ship Interdiction System
NSM	Naval Strike Missile
OTH	Over-the-Horizon
PAFMM	People's Armed Forces Maritime Militia
PLA	People's Liberation Army
PLARF	People's Liberation Army Rocket Force
PPLI	Precise Participant Location Identification
PRC	People's Republic of China
PrSM	Precision Strike Missile
PWSA	Proliferated Warfighter Space Architecture
SAR	Synthetic Aperture Radar
SATCOM	Satellite Communications
SIF	Stand-In Forces
SIGINT	Signals Intelligence
SIGMAN	Signature Management
SPMA	Statistical Priority-Based Multiple Access
SRBM	Short-Range Ballistic Missile
TDMA	Time Division Multiple Access
TDOA	Time Difference of Arrival
TEL	Transporter Erector Launcher
TTNT	Tactical Targeting Network Technology
UAS	Unmanned Aircraft System
UAV	Unmanned Aerial Vehicle
UHF	Ultra High Frequency
USV	Unmanned Surface Vehicle
UUV	Unmanned Undersea Vehicle
VHF	Very High Frequency
WEZ	Weapons Engagement Zone

METHODOLOGY

This thesis is a qualitative, doctrine-led feasibility study of the Expeditionary Advanced Base Operations (EABO) concept, with specific focus on its distributed intelligence, surveillance, and reconnaissance (ISR) functions. It does not produce new empirical data; rather, it synthesizes open-source doctrinal publications, government reports, peer-reviewed scholarship, and credible defense reporting to render a structured judgment on whether EABO can perform its intended functions within the threat environment posed by the People's Republic of China (PRC) anti-access/area-denial (A2/AD) complex.

Research Question

The central question is: *To what extent can EABO units conduct persistent, distributed ISR and contribute to joint kill chains inside the PRC's weapons engagement zone (WEZ) given current and near-term sensor, communications, and logistics constraints?*

Two supporting questions support the central inquiry. First, what capability gaps in the EABO concept currently constrain its feasibility? Second, what combination of mitigations, alternatives, and complementary concepts might offset those gaps?

Analytical Framework

Feasibility is assessed across four dimensions adapted from the joint DOTMLPF-P construct: Operational/Tactical, Technical/C2, Logistical, and Human. Each dimension is scored as Low, Medium, or High feasibility against three stated criteria:

- *Achievability* — can the function be performed with current or near-term (programmed by 2030) capabilities?
- *Sustainability* — can the function be performed persistently, or only once?
- *Survivability* — can the force performing the function reasonably expect to survive long enough to repeat it?

A dimension scoring Low on any of the three criteria is rated Low overall; a dimension scoring Medium or higher on all three is rated Medium or High accordingly. This rubric is applied in Chapter 6.

Time Horizon

The study uses 2027–2030 as the planning horizon. This window reflects PRC capability benchmarks rather than a prediction of conflict intent. President Xi has directed that the People's Liberation Army (PLA) be capable of unification operations against Taiwan by 2027 (Office of the Secretary of Defense, 2024). Admiral Davidson's 2021 testimony to Congress framed the same period as the window within which the PLA could be capable of acting (Davidson, 2021). The window is therefore one of capability readiness, not of intent, and the thesis takes no position on the likelihood that conflict will occur within it.

Source Selection

Sources were prioritized in the following order: (1) U.S. and allied military doctrinal publications (Tentative Manual for EABO 2nd Edition, A Concept for Stand-In Forces, LOCE, JP 3-60, JP 3-85); (2) U.S. government primary sources (Congressional Research Service reports, OSD China Military Power Reports, Government Accountability Office assessments); (3) peer-reviewed academic and policy scholarship; (4) established defense think-tank work (CSIS, RAND, CSBA, CNAS, Jamestown Foundation, Project 2049); (5) credible defense trade reporting (USNI News, Breaking Defense, Defense News, Naval News, Janes); and (6) manufacturer technical specifications, which are noted where relied upon and treated with appropriate skepticism.

Scope Limitations and Assumptions

The study is restricted to open-source material. Several load-bearing claims, particularly regarding the vulnerability of low-probability-of-intercept and low-probability-of-detection (LPI/LPD) waveforms to AI-enabled signals intelligence, sit in a region where the unclassified literature is thin; these claims are flagged where they occur.

Three explicit assumptions are made. First, host-nation access to the First Island Chain (principally Philippine and Japanese territory) is assumed to be granted. This assumption is not trivial. Recent Philippine policy under the Marcos administration has made it more credible, but Japanese political will to host kinetic shooters on the Ryukyus remains contested. The assumption is made to isolate the tactical-operational feasibility question from diplomatic variables. Second, the conflict scenario assumed is a high-intensity, peer-on-peer engagement centered on a Taiwan contingency. Lower-intensity gray-zone scenarios are addressed only briefly. Third, the EABO concept is evaluated against the doctrine published in the 2023 second edition of the Tentative Manual. Emerging modifications to that doctrine are referenced where relevant but the second edition is the primary baseline.

Limitations of Wargame Evidence

The feasibility judgments in this thesis draw on the published findings of unclassified wargames, principally the CSIS *First Battle of the Next War* study (Cancian, Cancian, and Heginbotham, 2023). This evidence base is treated as indicative rather than definitive: wargames are sensitive to rule design, scenario assumptions, and the resolution at which space ISR and electromagnetic warfare are modeled. The CSIS study's base case excluded pre-positioned Marine Littoral Regiments (MLRs); an excursion that allowed pre-positioning produced meaningfully different results. Where the analysis here relies on wargame findings, the underlying assumptions are noted.

1. ANTI-ACCESS/AREA DENIAL (A2/AD) IN THE INDO-PACIFIC

1.1 Defining the A2/AD Concept

Anti-access/area denial, or A2/AD, describes a military strategy that employs various systems and capabilities with the goal of preventing or limiting an adversary's military forces from operating freely within a specific geographical area or theater. The methods employed in an A2/AD system aim to restrict access to key areas such as sea lanes and airspace. The development and deployment of A2/AD systems and capabilities by the PRC have become a key focal point in defense planning for the United States and its allies in the Indo-Pacific. Although it can be broadly defined, A2/AD strategy can be academically and doctrinally broken into two complementary components: anti-access (A2) and area denial (AD) (Kearn Jr., 2013).

It should be noted that “A2/AD” is a U.S. analytical category rather than a PLA self-description. PLA writings frame the same activity in terms of *counter-intervention*, *systems destruction warfare*, and *informatized* and *intelligentized* warfare (Engstrom, 2018). The distinction matters because systems destruction warfare seeks to collapse an adversary's operational system by attacking critical C2 nodes, ISR architectures, and logistics chains rather than attriting individual platforms — and EABO units, as nodes in the U.S. C5ISR-T architecture, are exactly the kind of target this approach is built to attack.

Anti-access systems and capabilities are employed with the intent to prevent entry into a specific region or theater. These involve longer-range capabilities such as ballistic missiles that can target adversary air and naval bases at great distance within the contested area. These systems also hold high value naval targets, such as aircraft carriers, at risk. The intent of these tactics and systems is to keep an adversary entirely outside of a specific region or force them to operate at a range of reduced effectiveness.

Area denial involves shorter-range systems and tactics that aim to limit freedom of maneuver within a contested area. Examples of area denial systems include short-range ballistic missiles, coastal defense cruise missiles, and integrated air defense networks. These systems intend to disrupt an adversary's operations within a specific region and limit their freedom of movement.

A2/AD complexes are not designed or employed to be an impenetrable boundary. Their purpose is to make operations in a specific geographic region so costly and complex that an adversary is deterred from intervening, and, if they do intervene, sufficiently limit their advance and force them to accept disproportionately high casualties. This strategy provides a means for a force that could be perceived as inferior to effectively offset the qualitative or quantitative advantages of a superior military power.

1.2 Historic Context and Strategic Intent

China's efforts to develop a robust A2/AD strategy have been a decades-long, focused endeavor to transform the strategic environment of the Western Pacific. The 1995-96 Taiwan Strait Crisis, when the United States deployed two carrier strike groups to the Western Pacific, was a significant catalyst for this development. In this event, the People's Liberation Army (PLA) struggled to effectively prevent or even reasonably contest foreign intervention in its near seas (Mahnken, 2022).

As a result of the exposure of these vulnerabilities, the PLA initiated a full-scale reevaluation of its defense posture, and redirected its focus to creating a specific anti-access capability designed to prevent a superior adversary, such as the United States, from establishing itself within the region and thus being able to build strength or establish a presence in the Western Pacific.

Ultimately, China's A2/AD strategy is designed to inhibit the U.S. ability to intervene in regional conflicts and, therefore, limit the U.S. military's ability to project power in the region. By applying measures that impose extreme cost and difficulty to military operations in areas critical to Chinese national interests, such as the Taiwan Strait and the South China Sea, China seeks to credibly deter U.S. and allied intervention in order to create a favorable military balance in the region. This strategy provides China with greater freedom of maneuver in its near seas while simultaneously preventing adversaries from operating in the region.

1.3 A2/AD Systems and Capabilities

The PRC's A2/AD system uses a layered defense strategy that incorporates an arsenal of kinetic capabilities enabled by non-kinetic systems and command architecture. This robust complex is a system of redundant command, control, communications, computers, cyber, intelligence, surveillance, reconnaissance, and targeting (C5ISR-T) assets which include ground-based over-the-horizon (OTH) radar, airborne early warning systems, and a broad space-based surveillance capability (Newman et al, 2022). These systems allow the PLA to combine sensor data in near-real time and pass high confidence target information to mobile launchers throughout the First and Second Island Chains.

The strategic deterrent provided through the combination of kinetic assets, consisting of intermediate and medium range ballistic missiles used for anti-access and shorter-range coastal defense cruise missiles utilized for area denial, significantly reduces the reach and effectiveness of traditional naval forces. These systems collectively establish a highly contested environment, in which even low-signature elements can be rapidly detected and engaged.

1.4 Geographical Focus

China's A2/AD strategy is structured as multiple concentric layers of geographical space with escalating levels of hostility that increase the burden for any potential adversary to enter the theater. The overall structure is designed to make the maritime domain immediately adjacent

to the Chinese mainland a hostile environment where the PLA has the ability to exercise command over both sea and air.

Geographically, the strategy encompasses the First and Second Island Chains, as well as the Taiwan Strait and the South China Sea. The strategic objectives in these regions combine to reinforce the overall purpose of the A2/AD system.

First Island Chain

The First Island Chain describes the string of archipelagos in the Western Pacific, ranging from the Kuril Islands, through Japan, Taiwan, and the Philippines, and extending to Borneo. This region is the inner core of the A2/AD complex, where the PRC has the greatest ability to project its power. The PLA has employed fortified missile systems and developed naval capabilities with the goal of exercising sea supremacy in the region (Krepinevich, 2015).

Second Island Chain

Stretching from the Ogasawara Islands through Guam to western New Guinea, the Second Island Chain serves as the outer layer of strategic planning in the Western Pacific (Krepinevich, 2015). While the U.S. has key strategic hubs in this region, including Guam and Tinian in the Marianas Islands, the PRC aims to minimize the effectiveness of these resources. Specifically, the PLA intends to hold these critical hubs at risk using its arsenal of IRBMs.

Militarized Artificial Islands and Reefs in the SCS

Developments in the South China Sea represent a unique application of A2/AD in which the PRC has developed distributed strongholds in nearby regional maritime territory. By building and fortifying man-made islands in the Paracel and Spratly Islands, China has effectively extended its sensing and shooting capabilities well beyond the natural confines of its coastline. These artificial features provide the necessary infrastructure for China to deploy long-range radar, runways, and missile silos, thereby extending the reach of their A2/AD capabilities (Lake, 2023). This extends China's military power throughout Southeast Asia and challenges the freedom of navigation and operational security of other regional maritime forces.

Taiwan Strait

As a likely flashpoint for major conflict in the Indo-Pacific, the Taiwan Strait is the geographical and strategic center of China's A2/AD strategy. The PLA's arsenal of ballistic and cruise missiles holds land and maritime targets at risk. In addition, their IAMD system and naval forces significantly threaten any outside military interference in the region. These systems, designed to delay or prevent U.S. intervention, form a multi-layered defensive zone around the island of Taiwan and the Taiwan Strait.

2. EXPEDITIONARY ADVANCED BASE OPERATIONS (EABO)

2.1 Doctrinal Background for Distributed Operations

A doctrinal shift to distributed operations is an immediate response to the increased lethality and reach of A2/AD systems from adversaries. The U.S. military has sought to make its forces less centralized and therefore less vulnerable to targeting by an enemy by creating a more dispersed, mobile, and networked force. The goal of the dispersed forces is to effectively conduct offensive operations while making it more difficult for an enemy to locate and engage targets. As the fundamental, multi-service doctrine for countering the challenge of A2/AD, the U.S. Joint Concept for Access and Maneuver in the Global Commons (JAM-GC) integrates capabilities from all domains to ensure that the joint force maintains freedom of action in contested environments (Hutchens et al, 2017).

The U.S. Air Force supports this strategy through Agile Combat Employment (ACE), which aims to achieve resilience through a hub-and-spoke model to reduce dependence on large, fixed air bases. Distributed Maritime Operations (DMO), the Navy's model for a less centralized force, focuses on distributing sensors and shooters across a multitude of platforms to deny the adversary a clear focal point to attack. Littoral Operations in a Contested Environment (LOCE) is the strategic framework for integrating Navy and Marine Corps operations to leverage the land-sea interface as a position of tactical advantage to achieve sea control and facilitate sea denial (USMC, 2017). EABO is the tactical implementation of the LOCE framework. Specifically, EABO involves the deployment of mobile, low-signature Marine Corps units in stand-in locations within the adversary's WEZ. These small, mobile teams operate from austere advanced bases to provide critical functions such as forward sensing for targeting and long-range precision strikes against enemy assets (USMC, 2023).

2.2 EABO Structure: The Marine Littoral Regiment

The Marine Littoral Regiment (MLR) is a component of the Marine Corps and is most likely to employ the tactics of EABO. In the context of EABO, the MLR will serve as the stand-in force within the enemy WEZ (USMC, 2023). The MLR is designed to maneuver within the WEZ and conduct sea denial operations in order to enable fleet objectives. Key objectives of the MLR with regard to EABO include conducting surveillance and reconnaissance, denying or controlling key maritime terrain, conducting surface warfare operations, conducting air and missile defense, conducting strike operations, and conducting sustainment operations.

The MLR is composed of the Littoral Combat Team, the Littoral Anti-Air Battalion, and the Littoral Logistics Battalion (Feickert, 2025).

- **Littoral Combat Team (LCT):** The LCT is designed based on the infantry battalion structure and is intended to command EABO to facilitate sea denial. The LCT integrates reconnaissance and strike capabilities to enable the MLR to control critical maritime environments.

- **Littoral Anti-Air Battalion (LAAB):** Tasked with managing and directing air operations, the LAAB conducts air support as well as air and missile defense. The LAAB is designed to work in close coordination with other dispersed aviation units and C2 agencies.
- **Littoral Logistics Battalion (LLB):** The logistics battalion is tasked with supporting the distributed units by providing necessary tactical logistics. The LLB manages diverse missions, from littoral transportation to health services, necessary to sustain the MLR forces and ensure they remain operationally effective.

By creating a distributed network of EABs, the MLR makes the targeting cycle of the adversary harder to manage, as they are forced to use excessive ISR resources to locate the dispersed, low-signature forces. In addition, the MLR creates the link between sensors and shooters in the littoral zones through organic sensing and the feeding of data into the common operational picture (COP). This capability enables the execution of long-range precision fires. Ultimately, the persistent presence of the MLR, dispersed in the EABO construct, increases the costs of the adversary operating in the littoral space and supports friendly maneuver.

2.3 Contribution to Joint Kill Chains

The main purpose of EABO is to enable the F2T2EA kill chain — Find, Fix, Track, Target, Engage, and Assess (USMC, 2023). In high-intensity combat, the outside force (aircraft carriers, bombers) will likely have limited organic detection systems because they are operating at ranges far beyond their sensor reach. EABO forces serve as the stand-in force inside the enemy's WEZ, providing this capability (USMC, 2021). EABO units find enemy assets using ESM, small UAS, and visual observation; fix targets via ELINT triangulation or laser ranging; track and hand off contacts as needed; nominate and prioritize targets through sensor fusion; engage either organically (NMESIS, HIMARS) or by cueing joint shooters (a DDG firing a Tomahawk, a bomber launching LRASM); and perform battle damage assessment to close the loop (Joint Chiefs of Staff, 2024).

2.4 The Force Design 2030 Debate

EABO does not exist in a vacuum. It is the principal warfighting concept of Force Design 2030 (FD2030), the Marine Corps' decade-long modernization initiative initiated by then-Commandant General David Berger in 2020 and continued by his successors. FD2030 directs the divestment of legacy capabilities — all of the Corps' tanks, significant cannon artillery, most rotary-wing assault aviation, and a portion of infantry end-strength — and corresponding investment in long-range precision fires, unmanned systems, and the Marine Littoral Regiments that execute EABO (USMC, 2025).

This redirection has provoked the most consequential intra-service doctrinal debate the Marine Corps has experienced in a generation. A coalition of retired senior Marines — including former Commandants Charles Krulak and Anthony Zinni, retired Lieutenant Generals Paul

Van Riper and Greg Newbold, and others publishing through the *Marine Corps Compass Points* platform — has argued that FD2030 trades a combined-arms expeditionary force for a regionally-fixed, defensively-oriented missile force. Van Riper has argued that without tanks and sufficient cannon artillery the Marine Corps will be incapable of providing fleet marine forces with effective combined arms; Zinni has argued that combined-arms maneuver is being subordinated to a fires-delivery role (Cancian, M.F., 2025). Defenders of FD2030 — Berger, current Commandant Smith, and Lieutenant General Karsten Heckl — counter that the divestments target capabilities of marginal utility in a contested maritime environment where joint air and maritime supremacy cannot be assumed, and that the alternative to EABO is not a more lethal Marine Corps but an irrelevant one in the theater that matters most (Cancian, M.F., 2025; USMC, 2025).

For the purposes of this thesis, the Force Design debate is significant in two ways. First, the retired-officer critique is itself an authoritative open-source argument that EABO is not feasible — an argument made by Marines who have commanded at the highest levels of the institution being redesigned. Their concern with combined-arms attrition broadly overlaps with the feasibility conclusions reached here, though from a different analytical starting point. Second, the existence of the debate is a reminder that EABO's adoption was contested, and that the assumptions on which it rests remain disputed. The feasibility question is therefore not only one of capabilities and timelines, but also of whether the institutional commitment to EABO is durable enough to fund the contested-logistics and counter-ISR investments that the concept requires to succeed.

2.5 Challenges Facing EABO

EABO requires a shift in how operational tasks are accomplished relative to prior power projection doctrines. Instead of a transient, high-profile force deployed within an enemy's WEZ, EABO will utilize a persistent, low-signature stand-in force. Although the doctrine provides the joint force a powerful deterrent against A2/AD strategies, it is restricted by significant challenges for command and control, logistics, and survivability for Navy, Marine Corps, and other joint force organizations.

Operational and Tactical

At the tactical and operational level, EABO is challenged by survivability — specifically by requirements for emissions control (EMCON) and signature management (SIGMAN) — and by friction in the command and control (C2) process. In order to survive within the WEZ of the A2/AD complex, EABO units need to avoid counter-detection and targeting. To do this, they must significantly reduce and strictly manage their emissions. They also need to prioritize movement and relocation of forces to avoid targeting. If tasked with providing ISR and targeting support, the restraints of EMCON and SIGMAN can severely reduce the effectiveness of dispersed sensors or limit their ability to communicate data. In addition, the distributed nature of the units makes the coordination of assets to achieve specific objectives increasingly

difficult. In short, EABO forces are faced with a “transmit and die” paradox: if they do not emit, they fail to achieve their objectives and are tactically irrelevant; but if they do emit, they are extremely vulnerable to being targeted and destroyed (Clark, Patt, and Schramm, 2020).

Logistical and Human

Sustaining operations of the distributed model of EABO is likely to be a significant vulnerability. The same systems that threaten high-value naval assets and EABO units themselves pose a significant threat to any logistics assets. In order to sustain distributed units in the contested littoral region, logistics units must transition from the legacy model of large logistic transports to an integrated web of smaller, low-footprint, flexible connectors. Providing the necessary resources for the dispersed units will require integration of small, shallow-draft supply vessels such as the Landing Ship Medium (LSM). These will need to be supplemented by air logistics assets (helicopters, tilt-rotors, UAS) to ensure the EABO units have the resources they need to achieve their objectives. Medical resources will also be limited and will require detailed management to provide support to EABO units. Without the proper logistics infrastructure to support operations, distributed units will be unable to achieve their objectives or support joint operations.

3. DISTRIBUTED ISR SYSTEMS, CAPABILITIES, AND LIMITATIONS

In order to accomplish objectives and provide ISR and battlespace awareness, EABO units are tasked with employing a variety of systems. These systems include ground-based radars for air surveillance, UAS for persistent and longer-range reconnaissance, and passive sensors for SIGINT collection. In addition to the sensing capabilities, EABO forces employ counter-UAS systems and kinetic strike capabilities using mobile missile launchers.

3.1 Unmanned Aircraft Systems (UAS)

For the purposes of sensing and targeting, there exist a large range of UAS that are capable of contributing to EABO objectives. In order to maintain the small footprint required, however, some UAS are more practical for distributed operations.

To understand the different types of UAS that are available, they are broken down into 5 groups based generally on their size and operating characteristics. The table below breaks down the technical differences between different groups of UAS (Gettinger, 2024).

Group	Max Weight	Normal Altitude	Speed	Example Platforms
Group 1	< 20 lbs	< 1,200 ft AGL	< 100 knots	RQ-11 Raven, WASP, Puma
Group 2	21 – 55 lbs	< 3,500 ft AGL	< 250 knots	ScanEagle
Group 3	< 1,320 lbs	< 18,000 ft MSL	< 250 knots	RQ-7 Shadow, RQ-21 Blackjack
Group 4	> 1,320 lbs	< 18,000 ft MSL	Any Speed	MQ-1C Gray Eagle
Group 5	> 1,320 lbs	> 18,000 ft MSL	Any Speed	MQ-9 Reaper, RQ-4 Global Hawk

Although Group 4 and 5 UAS are the largest and generally most capable platforms, both groups face sustainment challenges in austere environments and remain vulnerable in high-threat A2/AD environments due to their larger radar signatures and logistical footprints, which contradict the EABO concepts of high mobility and low signature (Haffa and Datla, 2014). For these reasons, groups 1-3 are more suitable for EABO because they allow for organic sensing and targeting support but require significantly less of a footprint to operate.

Group 3 Systems

These systems provide the right tactical balance for expeditionary forces with increased endurance (usually 10-16 hours) and higher payload capacity than small UAS, but with size and weight suitable that they can be operated by a small team. These UAS benefit EABO operations through runway independent operation and modular payload bays to carry electro-optical/infrared (EO/IR) sensors, automatic identification system (AIS) receivers, and even small electronic warfare (EW) or kinetic attack packages.

System	Group	Launch / Recovery	Range	Endurance	Payloads
V-BAT ¹	3	True VTOL	130 km	10 hrs	EO/IR, SIGINT, Comms Relay
Jump 20 ²	3	Hybrid VTOL	185 km	13 hrs	EO/IR, SIGINT
RQ-21 Blackjack ³	3	Catapult / Skyhook	93 km	16 hrs	EO/IR, SIGINT, Comms Relay

¹ (Shield AI, 2025) ² (AeroVironment, 2025) ³ (Insitu, 2025)

The primary sensing tool for the MLR and stand-in forces of EABO will likely be the Group 3 UAS. Because they do not require specialized launch/recovery gear and because they can be launched from constrained areas, they can be deployed from the same types of remote, concealed locations as NMESIS and HIMARS. They can add value to EABO by conducting persistent, wide area search of the maritime domain and providing real-time targeting information to allow dispersed units to sense their immediate littoral environment without having to rely on satellite communications or far away strategic assets. However, Group 3 UAS will have to deal with constraints, such as line-of-sight (LOS) data link communications that can reveal the location of the EAB and reduced effective range as compared to Group 4 and 5 UAS (Gaines and Suh, 2023).

Group 1 & 2 Systems

Group 1 and 2 UAS compose the category of small UAS and are used at the organic/unit level with emphasis on portability and quick deployment. Group 1 UAS have a maximum weight of 20 pounds and operate below 1,200 feet. They are typically backpacked and hand-launched for short range reconnaissance. Group 2 UAS have a weight range of 21-55 pounds and operate up to 3,500 feet. Although Group 2 UAS require slightly greater logistics and support for operation, they provide more persistent, tactical ISR and maritime search capabilities required for larger units and dispersed EABO units.

System	Group	Launch / Recovery	Range	Endurance	Payloads
Flexrotor ¹	2	VTOL	120 km	12–14 hrs	EO/IR, SIGINT, Comms Relay
ScanEagle ²	2	Catapult/Skyhook	100 km	18–24 hrs	EO/IR, AIS, SIGINT
PDW C100 ³	1	VTOL	10 km	74–80 min	EO/IR; SIGINT; Comms Relay
RQ-11 Raven ⁴	1	Hand-Launch	10 km	60–90 min	EO/IR, Comms Relay
RQ-12 Wasp ⁴	1	Hand-Launch	5 km	50 min	EO/IR
RQ-20 Puma ⁴	1	Hand-Launch	20 km	3 hrs	EO/IR
Black Hornet 3 ⁵	1	Hand-Launch/VTOL	2 km	25 min	EO/IR

¹ (Airbus, 2025) ² (Insitu, 2025) ³ (PDW, 2025) ⁴ (AeroVironment, 2025) ⁵ (Teledyne, 2025)

Manufacturer specifications for the platforms above should be treated with caution; published figures are typically obtained under benign conditions and tend to revise downward against independent test data (Gettinger, 2019).

In the context of EABO, both Group 1 and Group 2 UAS are vital to establishing low-signature stand-in forces and to performing local maritime domain awareness and targeting functions. Both types of systems provide persistent tactical surveillance of littoral chokepoints and landing areas with limited logistics or electromagnetic (EM) footprints compared to larger strategic aircraft. Despite potential contributions to EABO, there are challenges to their employment in contested environments. The use of LOS data links makes them vulnerable to EW and jamming (Zyga, 2020). Additionally, the limited sensor payload and power output available due to the smaller size of the systems limit their ability to collect data. These smaller systems can potentially overwhelm enemy forces when employed in mass, but they are similarly susceptible to counter-UAS capabilities, weather, and kinetic air defenses.

Launched Effects / Loitering Munitions / One-Way Attack Drones

Loitering munitions are aerial munitions with built-in warheads designed to remain above a designated target area until a target is identified, at which point the operator commands terminal attack (Gettinger, 2019). Launched effects describes a broader family of systems launched from another platform — air-launched from a helicopter or drone, or ground-launched from a JLTV — to provide ISR, EW, or kinetic effects. Each provides a modular, networked means to extend the sensing and strike reach of distributed forces.

These systems offer EABO sea denial and sea control by providing organic, unit-level precision fires for small distributed units. They can independently target enemy radar sites, mobile missile launchers, and littoral surface vessels. Operated in networked swarms, they saturate enemy IADS and force expenditure of high-cost interceptors on low-cost, attritable drones. Limitations include small warheads inadequate for larger surface combatants, LOS data link vulnerability to EW and GPS denial, and shorter endurance than larger Group 3-5 UAS.

Empirical lessons from Ukraine

The first full peer-on-peer war fought under modern electronic warfare conditions has been the war in Ukraine, and its lessons for EABO are double-edged. Ukrainian forces have demonstrated that mass employment of cheap Group 1 and Group 2 UAS can impose disproportionate cost on a numerically superior adversary and provide persistent tactical ISR at scale (Watling and Reynolds, 2024). The same conflict has shown that attrition is severe: mean platform lifespans are measured in flights rather than months, and effectiveness shifts rapidly as both sides iterate on jamming and counter-UAS (Bendett, 2024). The implication for EABO is that the Group 1–3 UAS fleet must be procured with attrition rates one to two orders of magnitude higher than U.S. peacetime planning assumptions.

3.2 Ground-Based Surveillance Systems

Active Sensors

Active sensors emit energy to detect and track threats. The AN/TPS-80 Ground/Air Task Oriented Radar (G/ATOR) is an expeditionary, active electronically scanned array (AESA) multi-role radar system capable of detecting low-observable, low radar cross-section targets such as manned and unmanned aircraft systems, cruise missiles, and rockets (Feickert, 2025). As part of EABO, the G/ATOR acts as the main fire-control node for the MLR and integrates with the Common Aviation Command and Control System (CAC2S) to provide an integrated air picture. Active sensors provide the precision coordinates required for systems such as NMESIS and HIMARS to engage targets (Billard, 2025). However, because these systems continuously radiate energy, their emissions can be geolocated and the radars themselves engaged by enemy forces. Therefore, they are often operated in burst modes or tipped/cued by passive sensors to minimize their electromagnetic signature.

Passive Sensors and EW

Passive sensors enable EABO units to sense without being detected. Marine Corps radio battalions employ vehicle-mounted or man-portable SIGINT and EW suites to achieve this objective. These units collect intelligence from enemy communication transmissions and radar emissions to determine the locations of enemy ships and aircraft without transmitting or emitting any electromagnetic signal themselves (USMC, 2024). Additionally, ESM sensors are employed to characterize the signature of enemy radar systems, thus enabling commanders to identify the type of vessel based on emission parameters. The primary weakness of passive sensors is the dependency on enemy activity. If an enemy remains in EMCON, these sensors become ineffective.

3.3 Unmanned Surface and Undersea Vessels

The doctrinal treatment of EABO has historically been UAS-centric. The Ukrainian experience and the proliferation of low-cost maritime unmanned systems make this an increasingly unjustified omission; USVs and UUVs should be considered a fully equal pillar of distributed maritime ISR.

The Ukrainian Magura V5 USV — a 5.5-meter, approximately one-tonne autonomous craft with a unit cost reportedly between \$250,000 and \$300,000 — has, since its operational debut in May 2023, demonstrated effects no doctrinal forecast had assigned to a vessel of its size. By mid-2024, Magura V5 operations had reportedly sunk or damaged a meaningful fraction of the operational surface combatants of the Russian Black Sea Fleet, including the Tarantul-class missile corvette *Ivanovets* and the Ropucha-class landing ship *Tsezar Kunikov* in February 2024 (Adamopoulos, 2025). Subsequent variants (Magura V7) have demonstrated air-to-air missile launches from a surface platform, reportedly downing Russian Mi-8 helicopters and Su-30 fighters with modified R-73 and AIM-9 missiles. The cumulative effect has been to force the Russian Black Sea Fleet to relocate its operational warships from Sevastopol to

Novorossiysk, more than 300 kilometers further east — a strategic outcome achieved without a conventional surface navy.

The Ukrainian Sea Baby USV family has produced similar effects against fixed maritime infrastructure. In December 2025 Ukraine carried out what was reported as the first operational use of an attack UUV, the Sub Sea Baby, against a Russian Kilo-class submarine in Novorossiysk (Inside Unmanned Systems, 2025).

The relevance to EABO is direct. The First Island Chain is dense with the kind of littoral chokepoints — the Bashi Channel, the Luzon Strait, the Miyako Strait — where a USV/UUV capability proven against PLA Navy surface combatants would provide exactly the sea denial effect EABO is meant to enable, at a fraction of the cost and signature of a HIMARS or NMESIS battery. The U.S. Navy has begun procurement of comparable capability through USV Squadron Three, the GARC, and the DARPA NOMARS program (Eckstein, 2024), and the Replicator initiative is intended to deliver thousands of attritable autonomous systems (Hicks, 2023). For EABO specifically, the doctrinal centering of UAS as the primary distributed ISR pillar may need to be revisited: a Magura-class USV operated from a Philippine coastal hide site offers persistent maritime ISR and lethal effects with arguably better survivability characteristics than a Group 3 UAS, as it does not emit a radar signature and presents minimal thermal contrast against the sea surface.

3.4 Expeditionary Kinetic Weapons

The kinetic weapons underpinning the EABO framework enable the forces to extend their influence from land as a land-based adjunct to the naval fleet, and to project influence from the shore to create sea denial through power projection. A core system used to apply these kinetic capabilities is the NMESIS, which employs a JLTV to launch a Naval Strike Missile (NSM). The NSM is a subsonic, sea-skimming cruise missile, which has a greater than 100 nautical mile range, and includes an imaging infrared (IIR) seeker to allow for autonomous target recognition independent of active radar or GPS, providing high survivability in contested electromagnetic environments (Feickert, 2025). Additionally, EABO units could employ HIMARS, including the precision strike missile (PrSM). The PrSM incorporates a significantly increased range over previous missiles, with a range of 650 kilometers, and is expected to include future increments for moving maritime targets (Army Recognition, 2025). For even further reach, EABO units could employ the long-range fires launch (LRFL) to fire Tomahawk cruise missiles and the Maritime Strike Tomahawk (MST). The MST has an approximate range of 1,600 kilometers and can threaten larger surface combatants (DON, 2024).

Although these weapons provide EABs with lethality at the unit level, they are limited by numerous logistical and operational constraints defined by the EABO construct. The primary constraint to EABO is the logistics requirements associated with employing and sustaining long-range fires. The weapons that could be employed by these forces (Tomahawk, NSM, PrSM) require specialized equipment to transport and reload. This can prove difficult in

dispersed, austere locations where logistics support and resources are minimal. Also, although the launchers can be mobile, the F2T2EA process is robust and often requires significant electromagnetic signature for targeting and creates a large thermal and visual signature during engagement. These elements can be tracked and geolocated by enemy forces for counter-engagement, therefore requiring rapid repositioning following any fires. These kinetic systems also often rely heavily on targeting data from external sensors from air or ground assets. The sensors supporting the kill chain can be susceptible to targeting within the WEZ of enemy forces. Therefore, effective employment of these kinetic assets within EABO will require a balance between operating at stand-off range and signature management to maintain stealth within the A2/AD bubble.

4. COMMUNICATIONS, DATA LINKS, AND SIGNATURE MANAGEMENT

An isolated sensor in the First Island Chain will have minimal, if any, strategic significance if the information it provides is not communicated to a capable weapons system or command center. The move from linear kill chains to non-linear kill webs requires communications that are flexible, fault-tolerant, resilient, and connectable from a sensor to the most suitable shooter. In the contested electromagnetic spectrum of the Indo-Pacific, communications and data links will be regularly susceptible to degradation, denial, and jamming. The PRC's Strategic Support Force (SSF) — reorganized in April 2024 into the Information Support Force, Aerospace Force, and Cyberspace Force — has created a centralized apparatus for commanding space, cyber, and electronic warfare capabilities (Hoole, 2020; OSD, 2024). Any forces operating in this region, including EABO units, will have to maintain connectivity while being attacked by jamming, cyber attacks, and kinetic attacks to relay nodes.

Since regional limitations to communications will exist in the Indo-Pacific theater, an essential step for planning and conducting EABO is characterizing the electromagnetic operating environment. The limitations of a communication system include the relationship between frequency, range, bandwidth, and power that determine how effectively it can communicate information. For the purpose of EABO, any emissions can potentially lead to detection and engagement by adversary forces. Current SIGINT technologies allow for rapid detection, classification, and geolocation of radio emissions. The adversary can use a variety of ground-based direction-finding (DF) sites, airborne ELINT aircraft, such as Y-8 and Y-9, and space-based sensors to detect emissions. For this reason, the tactical concepts of signature vulnerability and jamming/denial apply specifically to EABO. Regarding signature vulnerability, even an individual Marine's radio transmission on an omnidirectional radio will provide a unique RF signature that can be geolocated, revealing the presence and location of the emitter. Even if the message itself is encrypted, the fact that the message was transmitted and where it originated allows for long-range strikes against the emitting unit.

4.1 Theory of Signature Management (SIGMAN)

The Marine Corps uses a comprehensive approach to SIGMAN which focuses on understanding your own signatures, recognizing the means an adversary can use to collect these signatures, and implementing countermeasures to cover or limit detection (USMC, 2023). In the context of EABO, this can manifest in tactics such as visual camouflage or radio silence. In the constant, active cycle of operations of opposing kill chains, SIGMAN focuses on three main components: understanding one's own signatures, identifying adversary collection methods, and taking action to mask signatures and limit detection. To be successful, the stand-in force must be able to understand the adversary kill chain and how to appropriately interrupt it. Various means of surveillance can be used to detect an EAB, and these different means can be fused together by the adversary to form a targeting solution. A full understanding of the different signatures that can be exploited is essential to the survivability of EABO forces.

Electromagnetic Signature

The EM signature is the most sensitive and vulnerable domain of intelligence information in the First Island Chain. This includes the emission of energy from communications equipment (HF, VHF, UHF, satellite uplinks) and active sensors (air defense radars, surface search radars). It also includes emissions from sources of electromagnetic interference such as power generators and vehicle electronics. Given the presence of PRC ELINT satellites, airborne DF sensors, and surface-based sensors, the EM signature provides the adversary with the opportunity to determine the position of a unit from hundreds of kilometers based solely on spectral fingerprint of a particular emitter, such as the G/ATOR or communication waveform.

Thermal and Infrared Signature

EABO forces produce thermal emissions through generator exhaust, vehicle engines, missile launches, and even cooking fires. Modern long-wave infrared (LWIR) sensors mounted on PRC EO/IR satellites, UAS, and aircraft can detect these emissions even through partial cloud cover. The transient but intense thermal signature of a missile launch is particularly difficult to conceal and provides an unambiguous detection cue.

Visual and Acoustic Signature

Although less precise than electromagnetic or thermal collection, visual signatures (vehicle tracks, antenna profiles, personnel movement) and acoustic signatures (engine noise, helicopter rotor wash) remain exploitable, particularly by the People's Armed Forces Maritime Militia (PAFMM) and PRC commercial satellite imagery providers conducting persistent overhead observation.

4.2 SIGMAN Process

The Marine Corps approach to SIGMAN comprises three iterative steps:

- **Signature Monitoring:** Commanders monitor for feedback on emissions produced by their units to avoid inadvertent exposure. Use of self-monitoring systems can identify accidental transmissions and prevent unnecessary emissions. Without a self-monitoring process, a unit may incorrectly believe it has managed its signature while it has mistakenly broadcasted distinct emissions (USMC, 2023).
- **Adversary Correlation:** Intelligence support must regularly analyze the adversary's collection patterns. This involves mapping of overflight of satellites and coverage areas of OTH radars. By correlating friendly maneuver with windows of reduced adversary coverage, commanders can time high-signature events to minimize detection probability (USMC, 2023).
- **Countermeasure Implementation:** Based on the assessment, EABO forces can use active and passive measures to reduce exposure. This ranges from strict EMCON protocols to deployment of multispectral decoys to dilute adversary targeting capabilities. The objective of these countermeasures is to manipulate the adversary's

perception of friendly forces and complicate their targeting of friendly forces (USMC, 2023).

4.3 Communication Networks and Tactical Data Links

The success of EABO, specifically with regard to communications, relies on establishing a “network of networks,” where specific communications and data link networks are selected based on the mission phase, tactical purpose, and electromagnetic threat level.

Link 16

As the most used standard for joint and coalition operations, Link 16 provides the primary means of developing the common tactical picture (CTP) among all units within an operation or geographical region. Link 16 uses UHF band (960-1215 MHz) and operates under a time division multiple access (TDMA) format which allocates a unit-specific timeslot for each user function (Joint Chiefs of Staff, 2015). In order to provide a measure of resilience when operating in contested environments, Link 16 systems use a frequency-hopping technology over 51 separate frequencies to avoid interference from jamming. Using Link 16, units are able to transmit local air and surface tracks to the larger network through J-series messages, thereby giving the fleet commander visibility of the same information that Marine Corps stand-in forces have. Additionally, Link 16 enables the dissemination of C2 commands and engagement orders and provides necessary blue-force tracking through PPLI.

Despite its multitude of advantages, there are several limitations to Link 16 that are relevant to EABO. Because Link 16 is a high-power, omni-directional transmitter, it is not a stealthy system. Persistent transmission can be detected by enemy ELINT or ESM. With regard to data, although Link 16 was designed to optimize track data, it does not have sufficient bandwidth to handle large volumes of data such as bulk ISR information involving high-definition video or unprocessed sensor measurements that enable advanced fusion. Finally, since Link 16 is a UHF waveform, it is primarily limited to LOS.

Tactical Targeting Network Technology (TTNT)

To meet the speed associated with modern naval warfare, the Navy and Marine Corps have used the technology known as Tactical Targeting Network Technology (TTNT). As an IP-based ad-hoc mesh network, TTNT is specifically engineered to address the bandwidth limits imposed by the existing Link 16 architecture and allows for timely exchange of information between assets. TTNT is engineered to operate at L-band frequencies and utilizes a statistical priority-based multiple access (SPMA) protocol for communications. For the purpose of EABO, TTNT forms the basis for advanced sensor fusion. With a throughput of up to 10 Mbps, TTNT is able to process high resolution images and transfer files much larger than those that could be transferred through Link 16 (Collins Aerospace, 2020).

However, the increased throughput capability comes at a cost in terms of signature management. Since TTNT is a wide band emitter that transmits at higher power settings, it

inherently carries a higher risk of detection. Although TTNT does contain low probability of intercept/low probability of detection (LPI/LPD) capabilities, it is generally less stealthy than directionally linked methods and will require careful consideration of how it will be employed in contested environments.

Cooperative Engagement Capability (CEC)

Cooperative Engagement Capability (CEC) is the enabling data link for Naval Integrated Fire Control - Counter Air (NIFC-CA) using a highly directional C-band data distribution system but is limited to LOS (DON, 2005). In the context of EABO, CEC provides essential fire control capabilities needed for cooperative engagements throughout a dispersed operating environment. The CEC enables destroyers and cruisers to launch missiles, such as SM-6, at targets provided by an EABO-based sensor, such as G/ATOR.

The Marine Corps and Navy systems described above are intended to plug into the broader joint architecture of Combined Joint All-Domain Command and Control (CJADC2), which provides the conceptual envelope for distributing sensor data across services. Marine-specific gateway solutions such as MANGL serve to translate between Marine Corps-organic networks and the larger joint enterprise.

4.4 EMCON Procedures and Pulsed Operations

In order to sustain survivability in hostile environments, the use of EMCON by stand-in forces enables them to employ a tiered structure of emissions silence (Tektronix, 2018). EMCON Alpha is total silence — no radar, no radio, no data links — with passive sensors only and no contribution to a common operational picture. EMCON Bravo permits limited essential communications, mostly receive-only with highly directional transmissions allowed. EMCON Charlie is a transitional state in which units may use mission-essential systems for limited periods to execute a specific tactical task such as a fire mission or critical targeting handoff. EMCON Delta has no restrictions; in the EABO context, sustained EMCON Delta is unlikely to be survivable.

The main operational problem that exists when attempting to operate in EMCON is the trade-off between hiding from the adversary and the need to communicate data to other units to provide targeting support. Doctrine addresses this problem by using pulsed operations. In this way, units operate in EMCON for certain time periods and only send data in short, high-bandwidth pulses when they need to. These pulses minimize the amount of time an adversary has to detect, locate, and effectively target the forces. Whether or not a unit will break EMCON is a high-risk, tactical decision. Decentralizing this decision is critical to enable unit commanders the ability to exploit fleeting windows of opportunity or to defend themselves without significant delay of centralized command processes.

4.5 Enabling Technologies and C2 Resilience

Managing emissions and reducing signatures is generally focused on procedural discipline, but there are technologies that provide tools to minimize risk associated with emissions in EABO. Using LPI/LPD waveforms can ensure required communications remain hidden or avoid detection. In order to avoid being a beacon while communicating, EABO forces use advanced waveforms designed to mimic background noise or complicate geolocation.

Advanced Waveforms and Directional Links

- **Direct Sequence Spread Spectrum (DSSS):** DSSS uses a pseudo-noise code to spread digital data over a large frequency band. The power spectral density of the signal is reduced, dropping below the noise floor of standard receivers. Without the specific spreading code, the transmission will only cause a minor increase in background noise, making detection very difficult. This is applied to satellite links, UAS controls, and data links (Stojanović et al, 2024).
- **Frequency Hopping Spread Spectrum (FHSS):** FHSS uses randomly changing frequencies, hundreds or thousands of times per second, according to a unique pseudo-random sequence. This hopping of frequencies makes it very difficult for enemy jammers to target or intercept the signal (Ristić et al, 2022).
- **Millimeter Wave and Directional Links:** High-bandwidth, low-signature communications, such as millimeter wave and directional links, can transmit significant amounts of data while limiting detection. Implementing these tactical networks reduces the likelihood of geolocation or jamming (Harvey et al, 2019).
- **Burst Transmissions and Data Mules:** Burst transmissions break a data stream into short packets, which prevents an adversary from maintaining a signal lock on the source of the signal and thus prevents geolocation (Sklar, 2001). Data mules use low-power, short-range communication between a stationary node and a passing UAS or USV. Data is transmitted via the data mule from a remote or elevated location, minimizing the likelihood of detection of the EABO unit (Palma et al, 2017).

Space-Based Transport

An emerging communication net that solves the LOS limitation is low earth orbit (LEO) transport. The Proliferated Warfighter Space Architecture (PWSA) employs a robust mesh network in space that uses inter-satellite links to maintain constant connection. By operating in LEO, these systems offer significantly reduced latency compared to geosynchronous satellites and provide high-speed, real time data exchange in contested environments. These proliferated LEO satellites can enable Ka-band uplink and downlink for mission data and C2 as well as offer a space-based Link 16 connection (SDA, 2023). The dependency risk should be noted: even if PWSA delivers on schedule, EABO units will be transmitting to LEO satellites that are themselves targetable by the PLA's growing counterspace arsenal of co-orbital ASATs, direct-ascent missiles such as the DN-series, and electronic and cyber means (Stokes et al, 2020).

Data Interoperability and C2 Networks

As the Marine Corps transitions from hardware-based systems to a software-based and data-driven model for managing network architecture, the interoperability of data transfer and C2 networks is critical to mission success. Integration of black core architecture, which encrypts all data at the same level before it is transmitted through the transport layer, allows multiple networks to transmit the data, regardless of the inherent trust of the network (Rose et al, 2020). Therefore, EABO units can send sensitive targeting data over military satellite, commercial 5G, fiber optic, or tactical radio mesh. Additionally, this architecture provides significant resiliency; if an adversary jams a primary communications link, the systems can automatically route through a secondary link.

Gateway systems such as the MAGTF Agile Network Gateway Link (MANGL) system act as a universal translator for Navy and Marine Corps assets. The MANGL system accepts data from various sources (Link 16, TTNT) and then transmits the data in a format acceptable to the intended recipient (Laird, 2020). In addition to the connectivity provided by MANGL, the CAC2S Small Form Factor (SFF) variant condenses the capabilities of a traditional direct air support center into a couple pelican cases of gear (ALSSA, 2025). This significant reduction in logistics footprint enables small, dispersed EABO units to create a complete tactical picture and conduct fire coordination with a reduced electromagnetic signature and increased mobility.

5. THREATS TO EABO

The PRC has developed an effective and redundant system that is capable of creating a costly operating environment for any opponent in the Western Pacific. The PRC uses its arsenal of kinetic capabilities to form multiple layers of defense and employs a robust strategy of deterrence and denial. The development and employment of a wide array of ballistic and cruise missiles, hypersonic technologies, a large fleet of naval ships, and integrated air defenses are all part of a plan to negate the power projection advantages of the U.S. and its coalition forces.

5.1 C5ISR-T Architecture and Space-Based Surveillance

China's A2/AD system relies heavily on its highly advanced C5ISR-T architecture, which functions as the nervous system for its kinetic weapons. This architecture ties together land-based OTH radar systems, airborne early warning systems such as the KJ-500/600, and maritime reconnaissance platforms to provide a unified COP for all participating elements of the system. The C5ISR-T architecture provides the PLA with the capability to rapidly integrate sensor data from various platforms and allow commanders to quickly transfer high-confidence targeting information from remote sensors to mobile missile launchers in near real-time.

The threat of space-based surveillance poses the most pervasive and technically challenging problem for EABO survivability. A large constellation of space-based platforms provides the large-scale area surveillance required to monitor the vast expanse of the Pacific Ocean. Utilizing a combination of electro-optical satellites for identification, SAR to penetrate through clouds and identify targets, and ELINT to detect radio emission of carrier strike groups and other dispersed units, China maintains track of moving targets hundreds of kilometers off its coast. The PRC operates a technologically sophisticated, multi-layered space surveillance architecture consisting of more than 510 ISR-capable satellites to monitor the First Island Chain continuously (SatNews, 2025).

The Yaogan series consists of EO, SAR, and ELINT satellites, developed to be used together to identify and locate mobile targets across the Indo-Pacific region. The Yaogan-41 satellite exists in a geostationary orbit, allowing it to maintain a persistent oversight of the region. The Yaogan-41 is assessed to be used for detection of thermal and optical signatures of surface connectors, logistics convoys, and missile launches in the First Island Chain. Yaogan-41 is primarily a tipping and cueing asset; it identifies potential activity that would require higher fidelity and cues higher-resolution LEO satellites to determine target information (Krebs, 2025).

Yaogan-29, Yaogan-33, and Gaofen-3 are examples of SAR satellites used by the PLA to compensate for environmental challenges of EO satellites, specifically persistent cloud cover and lack of sunlight. The SAR satellites use an active radar pulse to take images of the earth with resolutions of less than 5 meters (Krebs, 2025). In addition to high-resolution imaging, the PLA also uses a large number of ELINT triplets such as the Yaogan-30 and Yaogan-31

series (Krebs, 2025). The ELINT triplets operate in close proximity and use time difference of arrival (TDOA) to geolocate electronic emissions, locating radio signals such as shipboard radar systems, Link 16 tactical data links, and SATCOM uplinks (Stokes et al, 2020). The Yaogan-30 constellation provides a revisit rate of the South China Sea on the order of 30 times per day; given the continued growth of the constellation since the underlying source data was compiled, this figure should be treated as a conservative floor. Even brief radio transmissions using LPI waveforms are therefore at high risk of detection unless the waveform is completely masked and the transmission duration is insufficient for intercept.

China's strategy of civil-military fusion is an additional component of the military architecture of space-based assets. The Jilin-1 commercial imaging constellation is the world's largest sub-meter commercial imaging constellation, with a global revisit rate of approximately 38-40 times per day. Passing over the First Island Chain every 15 to 30 minutes, Jilin-1 has an advanced capability to capture high-definition video, allowing for tracking of moving vehicles and collection of speed and direction data for dynamic targeting (Stokes et al, 2020). Recent open-source reporting indicates that Jilin-1 has demonstrated the ability to track fifth-generation stealth aircraft through cloud cover (International Affairs Review, 2025), illustrating the rapid maturation of the constellation's capability against difficult targets.

The PLA is rapidly developing AI and ML to manage the vast amount of data generated by their satellite constellations. AI/ML-based automatic target recognition algorithms can quickly scan millions of images taken by satellites and identify anomalies that may not immediately appear to human analysts, such as new areas of cleared vegetation or the size and shape of specific vehicles (Barker, 2021). Through pattern of life analysis, AI models can identify even small changes in the environment, including variations in heat signatures or changes in vegetation density. Ultimately, the persistent and autonomous nature of the AI/ML-based processing makes it very difficult to maintain a low signature for EABO.

5.2 Airborne and Maritime Sensing

Below the layer of satellites, the PLA has developed numerous airborne platforms and sensors for the purpose of tracking and targeting its enemies in littoral zones of the First Island Chain. A mixture of reconnaissance and airborne early warning aircraft are employed to actively locate, track, and target any adversary forces (OSD, 2024).

EABO employment and survivability depend on the sea and the use of the maritime environment to move, insert, and sustain operations. A central component of the PLA's full-spectrum maritime sensing system is the People's Armed Forces Maritime Militia (PAFMM). This state-sponsored force of civilian vessels uses enhanced sensors and communications to act as a ubiquitous and low-observable sensor network that operates in the gray zone of conflict (Poling, 2019). In an EABO environment, PAFMM vessels can create a flood of sensors, report disposition of enemy forces, and identify targets.

Below the surface, China has established an expansive underwater sensor network commonly referred to as the “great underwater wall.” This system consists of fixed sonar arrays placed on the ocean bottom that monitor the acoustic signature of submarines and surface vessels (Pradhan, 2025). Floating platforms with radar, EO/IR cameras, and communications relay capabilities complement the seabed sensors, providing real-time detection capability regarding the transit of amphibious ships and supply vessels. Studies indicate that PLA sensors have been able to increasingly detect and classify noise generated by small vessels, potentially exposing covert nighttime insertion missions.

5.3 Kinetic Strike Complex

The sensing and detection methods of the PLA are the enablers for the robust kinetic strike complex that exists to attack targets with speed and volume. A key component of this threat is the PLA Rocket Force, which possesses a large inventory of ground-launched missiles. The Dong Feng-21 (DF-21) family, particularly the DF-21D “carrier killer,” provides anti-ship ballistic missile capability with ranges in the 1,500–2,150 km class, while the Dong Feng-26 (DF-26) intermediate-range ballistic missile extends that reach to approximately 3,000–4,000 km, threatening U.S. strategic hubs at Guam and major bases in Japan (Romero Meraner, 2023; Missile Threat, 2024). The Dong Feng-17 (DF-17) introduces a hypersonic glide vehicle, the DF-ZF, which uses aerodynamic forces to alter direction during high-speed atmospheric flight, complicating intercept by reducing decision time available to missile defenses (Missile Threat, 2024). Although recent assessments indicate the DF-17/DF-ZF combination is more likely tasked against fixed theater targets than against ships, its presence in the PRC inventory adds to the targeting problem facing distributed forces.

Coastal defense and cruise missile capabilities further densify the threat picture. The YJ-12 ramjet-powered ASCM with an air-launch capability reaches Mach 2–4 at ranges up to 460 km; the YJ-18 vertical-launch dual-mode missile extends to roughly 540 km; both incorporate terminal phase maneuvering to defeat naval defenses. The longer-range, subsonic YJ-62 is frequently deployed on transporter-erector-launchers to provide mobile, persistent threat to naval assets in the First Island Chain (Missile Threat, 2024).

5.4 Cyber, Electronic, and Counter-Space Threats

The kinetic threats described above are necessary but not sufficient to defeat EABO. The faster and more consequential threats to a distributed force may be in the cyber and electromagnetic domains. The PLA's Information Support Force, Aerospace Force, and Cyberspace Force — established in April 2024 from the restructured Strategic Support Force — concentrate the capability for offensive cyber operations against U.S. command-and-control networks, jamming and deception against tactical data links, and counter-space operations against the U.S. ISR and communications satellites on which EABO depends (OSD, 2024). For EABO specifically, the relevant threats include jamming of Link 16 and TTNT, GPS denial in the First Island Chain, cyber attack against MANGL or PWSA ground stations, and counter-space attack

on LEO transport and ISR assets. These threats are non-kinetic, can be employed below the threshold of armed conflict, and can degrade EABO well before a single missile is fired.

6. FEASIBILITY ASSESSMENT

This chapter applies the feasibility rubric introduced in the Methodology to render a structured judgment on EABO. Each of the four dimensions — Operational/Tactical, Technical/C2, Logistical, and Human — is assessed against the three criteria of Achievability, Sustainability, and Survivability, and rated Low, Medium, or High. A summary table is presented at the end of the chapter.

The temporal frame is the 2027–2030 capability window. As established in the Methodology, this is a window of PLA capability readiness derived from Xi's directive to the PLA to be capable of unification operations by 2027 (OSD, 2024) and Admiral Davidson's 2021 testimony on the same period (Davidson, 2021); it is not a forecast of conflict intent.

6.1 Operational and Tactical Feasibility

Achievability. The EABO concept is achievable in narrow tactical terms. The Marine Corps has fielded a first-generation MLR at 3rd MLR in Hawaii, with 12th MLR following in Okinawa and 4th MLR programmed (USMC, 2025). NMESIS batteries are operationally deployed; G/ATOR is in service; HIMARS is mature. The doctrinal and equipment foundations to conduct EABO at brigade scale exist. **Achievability: Medium.**

Sustainability. Sustaining EABO at scale is more difficult. High-fidelity wargames, principally the CSIS *First Battle of the Next War* study, indicate that MLR units suffer very high attrition rates once they initiate fires or transmit above certain thresholds (Cancian et al, 2023). The electromagnetic and thermal signatures of fires draw rapid PLA counter-fire, and the restricted geography of the First Island Chain limits off-road movement to predictable road networks. The Ukrainian artillery experience reinforces this: Ukrainian forces have demonstrated that shoot-and-scoot tactics work, but only at the cost of constant relocation, very short emission windows, and an equipment loss rate that exceeds any plausible U.S. peacetime replacement plan (Watling and Reynolds, 2024). **Sustainability: Low.**

Survivability. The survivability of an EABO unit is a function of three variables: the PLA's ability to detect it, the PLA's ability to fix and target it within its own kill-chain timeline, and the unit's ability to relocate faster than the PLA can close that loop. The first variable is moving against the U.S. force: the 510+ ISR-capable satellites in PRC inventory, combined with airborne and maritime sensing, make sustained concealment of a battalion-sized force increasingly implausible. The second is also moving against the U.S. force: hypersonic and ballistic strike timelines compress the relocation window. The third — relocation speed — is constrained by the physical limits of moving heavy systems like NMESIS and HIMARS off limited road networks under hostile drone observation. **Survivability: Low.**

Operational/Tactical Feasibility: Low.

6.2 Technical and C2 Feasibility

Achievability. The technical architecture for EABO — Link 16, TTNT, CEC, MANGL, the planned PWSA transport layer — exists or is programmed. CAC2S SFF reduces the C2 footprint to a manageable size for a distributed force. **Achievability: Medium.**

Sustainability. Sustained C2 within the WEZ depends on resilient transport across multiple paths. The “network of networks” concept is sound, but its sustainability is bounded by the survivability of the underlying transport. Persistent jamming of UHF/L-band, GPS denial, and saturation of LEO uplinks degrade the system faster than redundancy can restore it. The honest position is that the system will work intermittently, in pulses, and not as a continuous capability. **Sustainability: Low to Medium.**

Survivability. The “transmit and die” paradox is the core technical problem (Clark, Patt, and Schramm, 2020). LPI and LPD waveforms were engineered for an earlier generation of SIGINT processing. Modern PLA collection — particularly the ELINT triplets using TDOA geolocation combined with AI-assisted signal classification — is, on the best open-source reading, narrowing the gap. The thesis here flags this as a load-bearing claim that sits at the edge of the unclassified literature: definitive technical assessment requires classified data not available here. The directional conclusion — that the historical assumption of LPI/LPD as a survivability solution is eroding — is well supported in the unclassified work. **Survivability: Low.**

Technical/C2 Feasibility: Low.

6.3 Logistical Feasibility

Achievability. The contested-logistics platform on which EABO is doctrinally premised is the Landing Ship Medium (LSM). The Navy in late 2024 cancelled its initial industry solicitation due to bids exceeding budget expectations, and the lead-ship delivery target — originally April 2026 — has slipped to February 2029 at earliest (O'Rourke, 2025). Until then, EABO logistics depend on the legacy amphibious fleet, which is exactly the kind of large-signature target the EABO concept was designed to make unnecessary. **Achievability: Low.**

Sustainability. The Marine Corps is moving away from centralized logistics to support distributed forces, but the underlying demand — fuel, water, missiles, food, batteries — does not shrink. Distributed reloading of NMESIS, HIMARS, and Tomahawk launchers requires specialized equipment and a logistics network that the LSM delay leaves unbuilt. Wargame results indicate that contested logistics assets are themselves preferred PLA targets, and that the loss of a logistics line renders the dispersed force immobile (Cancian et al, 2023). **Sustainability: Low.**

Survivability. Logistics vessels operating in the WEZ face the same ISR-strike complex as the force they support, but without the small-signature advantages claimed for EABO units

themselves. Stern-landing ships and other interim solutions provide a stopgap but are not survivable platforms in a peer conflict. **Survivability: Low.**

Logistical Feasibility: Low.

6.4 Human Feasibility

Achievability. The Marine Corps has the personnel structure to field MLRs and is recruiting and training to the model. Initial operational capability has been achieved. **Achievability: Medium.**

Sustainability. Sustaining human performance under EABO conditions is the most understudied dimension of the concept. Casualty evacuation in contested littoral terrain is likely to be denied by PLA IADS, requiring prolonged casualty care by units with limited Role 1 and almost no Role 2 capability. Wounds that would be survivable in a permissive environment will become fatal. The cognitive and psychological burden of operating in small isolated teams under continuous overhead observation, without reliable resupply, evacuation, or rotation, is significant and is not adequately addressed in current doctrine. **Sustainability: Low.**

Survivability. Beyond the kinetic survivability addressed in Section 6.1, the question is whether forces can remain combat-effective through the duration of a conflict. Historical analogues from prolonged isolated operations (Bataan, Guadalcanal supply crises, Wake Island) suggest combat effectiveness degrades sharply after the first two to three weeks of disrupted sustainment. **Survivability: Low.**

Human Feasibility: Low.

6.5 Summary

Dimension	Achievability	Sustainability	Survivability	Overall
Operational/Tactical	Medium	Low	Low	Low
Technical/C2	Medium	Low-Medium	Low	Low
Logistical	Low	Low	Low	Low
Human	Medium	Low	Low	Low

The pattern is consistent: EABO is achievable as a concept and partially achievable as a fielded force, but is not sustainable or survivable at the scale and duration the doctrine envisions, against the threat the doctrine was designed to counter, within the 2027–2030 window. The conceptual objectives of EABO appear challenged by the practical realities of both the material and technical aspects of high-intensity peer conflict in the Indo-Pacific region unless the U.S. military establishes a more robust kill web and significantly improves the mobility and logistics resilience of its forces.

One important qualification should be entered against this summary judgment. The feasibility rubric used here evaluates EABO as a *persistent* concept — a force designed to remain inside the WEZ across the duration of a conflict. EABO may be substantially more feasible as a *one-shot* deterrence-by-denial instrument: a force that achieves significant kills in the opening 72 to 96 hours of a conflict before being attrited may still produce strategically decisive effects by shaping PLA decision-making, even if the units themselves do not survive. The CSIS wargame results are consistent with this reading. Measured against the one-shot standard rather than the persistent-presence standard, the operational and tactical dimension would arguably score Medium. This is the strongest version of the optimistic case for EABO.

7. MITIGATIONS AND ALTERNATIVE CONCEPTS

A feasibility assessment that ends at “Low” without engaging the available mitigations and alternatives is incomplete. This chapter examines the principal options the U.S. and its allies are pursuing to either close EABO's capability gaps or substitute for them.

7.1 The Replicator Initiative and Attributable Mass

The Replicator initiative, announced by then-Deputy Secretary of Defense Kathleen Hicks in August 2023, directed the procurement of thousands of “all-domain attributable autonomous” (ADA²) systems within 18 to 24 months, with the explicit purpose of offsetting PLA quantitative advantages with attributable mass (Hicks, 2023). The first Replicator capabilities reached warfighters in May 2024, and the program was on track to meet its August 2025 milestone (Defense Innovation Unit, 2025). For EABO, Replicator-class systems offer a partial substitute for the manned-unit ISR and fires functions: a swarm of attributable autonomous USVs, UUVs, and UAS can perform sensing and limited engagement functions without exposing Marines.

7.2 The Hellscape Concept

In June 2024 Admiral Samuel Paparo, commander of U.S. Indo-Pacific Command, publicly described his intention to “turn the Taiwan Strait into an unmanned hellscape” using a combination of unmanned air, surface, and subsurface systems to delay a PLA amphibious operation long enough for joint and allied forces to converge (Rogin, 2024). Hellscape is not a replacement for EABO; it is a complementary concept that targets the maritime crossing rather than the post-landing fight. But for the feasibility question, Hellscape is significant: it accepts that the manned stand-in force is likely to be attrited and substitutes attributable mass as the first line of denial. If Hellscape works, EABO does not need to be persistent — it needs only to be sufficient until the unmanned shield is established.

7.3 Allied and Partner Contributions

The thesis to this point has framed EABO as a U.S. concept executed by U.S. forces. This framing systematically understates the contribution available from allies and partners, every one of which alleviates the burden on EABO.

Japan's upgraded Type 12 surface-to-ship missile, with a reported range of approximately 1,000 km, completed development in December 2025 and is being deployed across the southwestern islands and home islands (Naval News, 2025). Deployed from the Ryukyus or Kyushu, the upgraded Type 12 covers much of the East China Sea and parts of the Chinese coast. A new Island Defense Surface-to-Ship Missile under development by Kawasaki Heavy Industries is reportedly targeted at a range on the order of 2,000 km. Japanese A2/AD-style coastal denial, executed by Japanese forces from Japanese territory, removes a meaningful share of the targeting problem from U.S. EABO units.

The Philippine Marine Corps unveiled its first operational BrahMos shore-based anti-ship missile battery in November 2025, deployed in Western Luzon within range of Scarborough Shoal. A second battery has been delivered, and a third was scheduled for delivery in early 2026 (Mangosing, 2025). With a 290 km range and Mach 3 terminal velocity, the BrahMos provides a credible sea-denial capability against PLA Navy operations in the South China Sea, executed by Philippine forces from Philippine territory.

Taiwan's Hsiung Feng III supersonic anti-ship missile and the longer-range Hsiung Feng IIE land-attack variant provide Taiwan-organic strike capability. Australia is acquiring SM-6 and Tomahawk to be fitted to its Hobart-class destroyers and acquiring the Naval Strike Missile for its surface fleet (Department of Defence Australia, 2024). The collective effect is that the targeting problem facing the PLA is no longer just U.S. EABO units; it is U.S. EABO units plus Japanese Type 12 batteries plus Philippine BrahMos batteries plus Taiwanese HF-III plus Australian SM-6 plus ROK Hyunmoo-III. A feasibility assessment of EABO that treats it as a stand-alone concept misses the synergy of the coalition picture.

7.4 Stand-Off Fires from Outside the WEZ

The case for EABO rests on the assumption that the outside force cannot reach into the WEZ without a stand-in sensor and shooter network. That assumption is being eroded by stand-off fires programs: the Air Force's LRASM and the developmental Hypersonic Attack Cruise Missile (HACM); the Army's Mid-Range Capability and Dark Eagle Long-Range Hypersonic Weapon; and the Navy's Maritime Strike Tomahawk all provide alternative pathways to engage maritime targets without putting Marines ashore. The penetration challenge shifts to the bomber or surface combatant launching the weapon, which the Air Force and Navy are addressing through their own stealth and standoff investments.

This is not a substitution argument; it is a portfolio argument. The relevant question for EABO feasibility is whether the marginal capability EABO provides — particularly the sensor function inside the WEZ — is worth the cost and risk relative to investing the same resources in penetrating ISR (such as long-range stealthy UAS) operating from outside the WEZ.

7.5 Counter-ISR

The single highest-leverage intervention to improve EABO feasibility is to degrade the PLA's ability to find EABO units in the first place. Counter-space operations against the Yaogan, Jilin, and TJS series; electronic warfare against PLA airborne ELINT and OTH radar; cyber operations against PLA ISR data fusion; and deception operations against the AI/ML processing pipeline all increase the time-to-detect and reduce the probability of correct classification. Of all the U.S. investment priorities the conclusion of this thesis will identify, counter-ISR has the largest direct effect on EABO survivability — because every additional minute of concealment is a minute in which the unit can complete its task and relocate.

8. CONCLUSION

The landscape of the Indo-Pacific is vast, and a potential conflict between the U.S. and its allies and the PRC would certainly see the use of unprecedented technologies, tactics, and operations. The potential for conflict certainly exists, with tensions in the South China Sea and the Strait of Taiwan rising day by day. It is incumbent on U.S. forces to understand the potential systems they will be operating and the tactics they will use to operate them.

The PRC has developed a robust, resilient, lethal A2/AD complex of systems designed to prevent or complicate foreign interference in the Western Pacific. To overcome the operational limitations imposed by this system, U.S. and allied forces must develop a strategy of distributed units to complicate adversary targeting and enhance coalition freedom of maneuver. For this reason, the EABO concept is tactically relevant and necessary. Dispersed units operating within the WEZ will enable the operations of the joint and coalition force.

However, when assessed against the structured feasibility rubric applied in Chapter 6, EABO scores Low across all four dimensions — operational, technical, logistical, and human — when evaluated against the standard of persistent presence and effects within the 2027–2030 capability window. The PLA ISR and targeting architecture, combined with the gap in contested logistics opened by the LSM delay, makes the persistent variant of EABO assessed as currently unfeasible. Survivability within the First Island Chain is dependent on strict adherence to EMCON and SIGMAN procedures, yet to complete their objective of contributing to targeting, EABO units are forced to emit. Despite enabling technologies such as LPI/LPD waveforms, advances in adversary detection systems significantly increase the risk of targeting by enemy forces. Critical logistics systems such as the LSM have not yet been fielded, and these logistics resources are similarly susceptible to the same targeting that EABO units face themselves.

The Force Design 2030 debate within the Marine Corps reflects the same underlying tension. The retired-officer critique has argued that the divestments required to field EABO have purchased a capability that may not survive the conditions for which it was designed; defenders of Force Design have responded that the alternative is irrelevance. The feasibility analysis here does not resolve that debate, but it does establish that the optimistic case for EABO depends on closing capability gaps — particularly in contested logistics and counter-ISR — that the current force-design timeline has not yet closed.

The mitigations and alternatives examined in Chapter 7 do not invalidate the Low feasibility judgment for the persistent-presence EABO concept, but they substantially change the picture for the broader portfolio of which EABO is one part. The Replicator initiative is delivering attritable mass on a schedule that may reach effect before EABO's logistics constraints are resolved. The Hellscape concept reframes the maritime denial problem so that EABO need only be sufficient until the unmanned shield is established. Japanese Type 12, Philippine BrahMos, Taiwanese HF-III, and Australian and Korean stand-off systems all impose part of

the targeting problem on the PLA without requiring U.S. Marines to be the principal stand-in force. And U.S. stand-off fires from outside the WEZ provide an alternative pathway to the same effects EABO is meant to produce.

To increase the operational feasibility of EABO, four prioritized advancements are recommended. First and most important, the U.S. military should invest in counter-ISR capabilities against the PLA space, airborne, and maritime sensing complex. Every additional minute of concealment compounds across the kill chain; this is the single highest-leverage intervention available. Second, the contested logistics gap must be closed as a matter of urgency — commercial designs, stern-landing ships, and unmanned logistics platforms can provide interim capability while LSM is brought into service. Third, kill-chain resiliency should be improved through further development of advanced C2 networks (PWSA, MANGL, JADC2) and through deliberate investment in C2 redundancy designed for graceful degradation rather than for clean operation. Fourth, tactical mobility for the stand-in force should be enhanced through smaller, more autonomous assets that reduce both the logistical footprint and the time required to relocate after each shoot-and-scoot cycle.

The success of EABO will ultimately be determined not by the lethality of individual shooters, but by the resilience of the network that connects them, the survivability of the logistics that sustains them, and the broader joint and allied portfolio inside which they operate. EABO is necessary but not sufficient. Sufficiency, on the basis of this analysis, is a function of investments not yet completed and concepts only now being fielded.

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